

## **Lab 4 - Operating System Tasks**

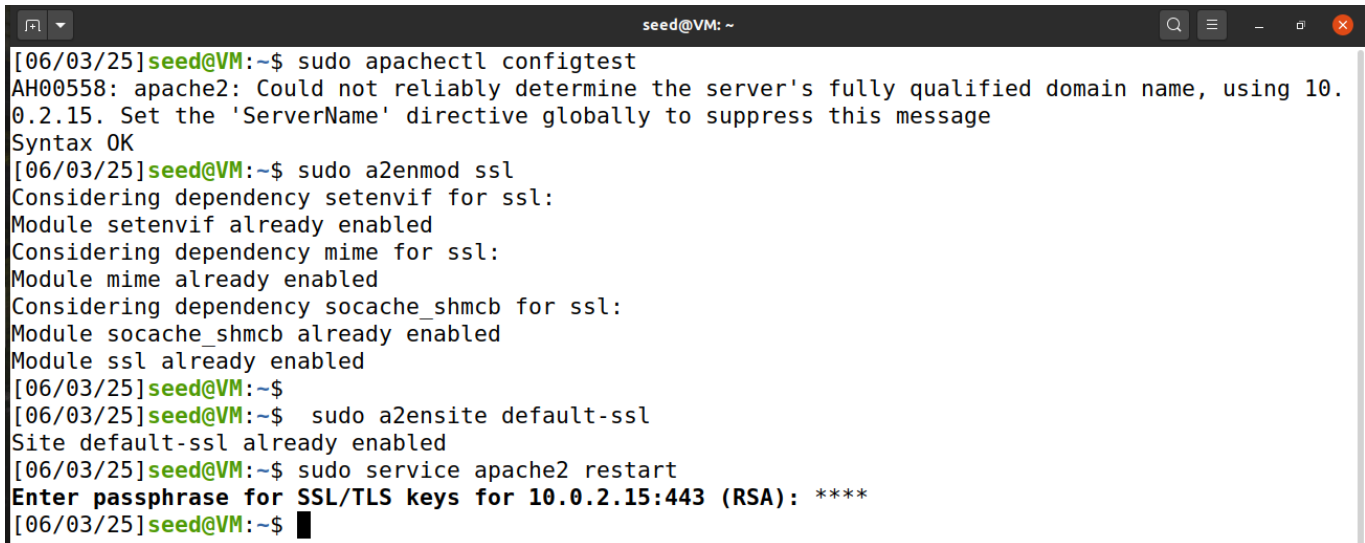
Name: Mohamed Ali Mohamed

SAP ID: 52755

Instructor: Sir Arsalaan

Course: Cryptography

## Task4 p1.2.png



```
[06/03/25]seed@VM:~$ sudo apachectl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.2.15. Set the 'ServerName' directive globally to suppress this message
Syntax OK
[06/03/25]seed@VM:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
[06/03/25]seed@VM:~$
[06/03/25]seed@VM:~$ sudo a2ensite default-ssl
Site default-ssl already enabled
[06/03/25]seed@VM:~$ sudo service apache2 restart
Enter passphrase for SSL/TLS keys for 10.0.2.15:443 (RSA): ****
[06/03/25]seed@VM:~$ █
```

Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task4 p1.2.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

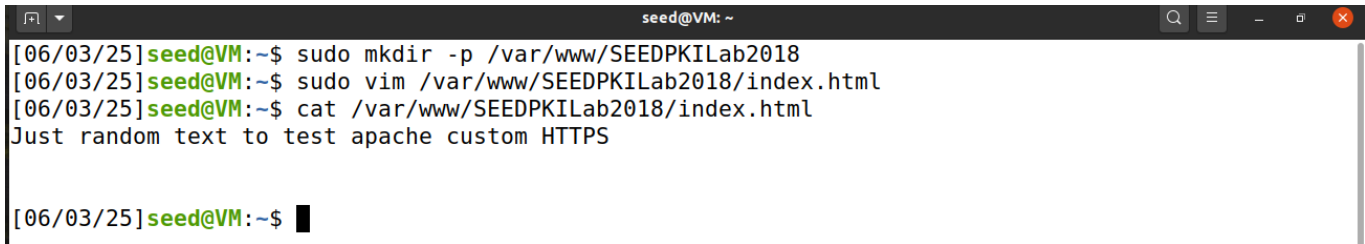
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## Task4 p1.3.png



```
seed@VM: ~  
[06/03/25]seed@VM:~$ sudo mkdir -p /var/www/SEEDPKILab2018  
[06/03/25]seed@VM:~$ sudo vim /var/www/SEEDPKILab2018/index.html  
[06/03/25]seed@VM:~$ cat /var/www/SEEDPKILab2018/index.html  
Just random text to test apache custom HTTPS  
  
[06/03/25]seed@VM:~$
```

### Decryption:

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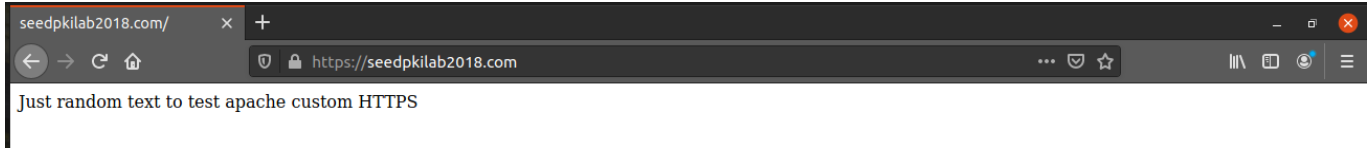
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## Task4 p1.4.png



### Decryption:

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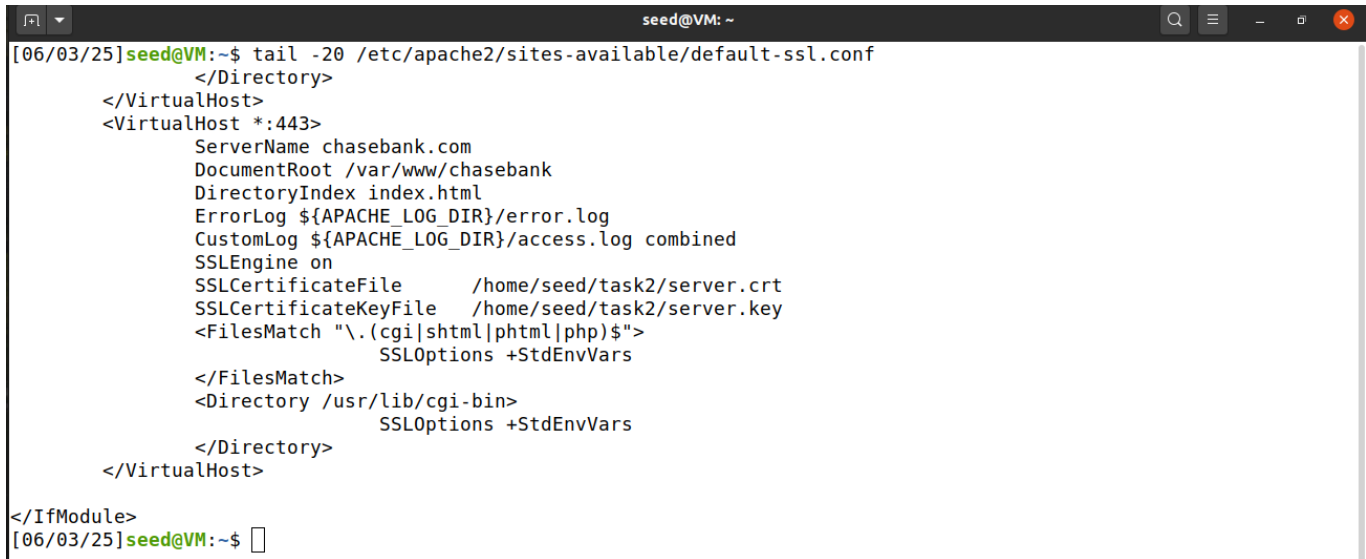
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## Task5 figure 1.1.png



```
seed@VM: ~  
[06/03/25] seed@VM:~$ tail -20 /etc/apache2/sites-available/default-ssl.conf  
        </Directory>  
    </VirtualHost>  
    <VirtualHost *:443>  
        ServerName chasebank.com  
        DocumentRoot /var/www/chasebank  
        DirectoryIndex index.html  
        ErrorLog ${APACHE_LOG_DIR}/error.log  
        CustomLog ${APACHE_LOG_DIR}/access.log combined  
        SSLEngine on  
        SSLCertificateFile      /home/seed/task2/server.crt  
        SSLCertificateKeyFile    /home/seed/task2/server.key  
        <FilesMatch "\.(cgi|shtml|phtml|php)$">  
            SSLOptions +StdEnvVars  
        </FilesMatch>  
        <Directory /usr/lib/cgi-bin>  
            SSLOptions +StdEnvVars  
        </Directory>  
    </VirtualHost>  
</IfModule>  
[06/03/25] seed@VM:~$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task5 figure 1.1.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

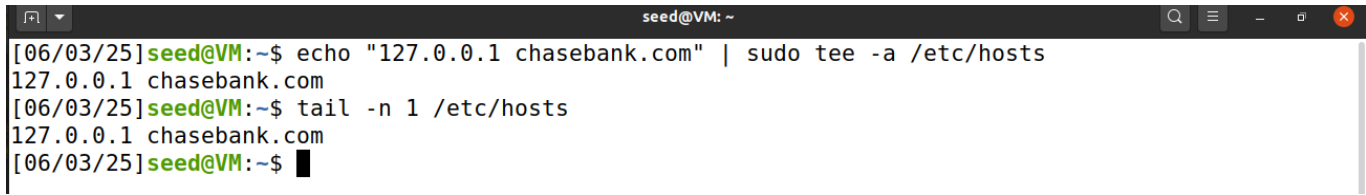
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## Task5 figure 1.2.png



```
seed@VM: ~  
[06/03/25]seed@VM:~$ echo "127.0.0.1 chasebank.com" | sudo tee -a /etc/hosts  
127.0.0.1 chasebank.com  
[06/03/25]seed@VM:~$ tail -n 1 /etc/hosts  
127.0.0.1 chasebank.com  
[06/03/25]seed@VM:~$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task5 figure 1.2.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

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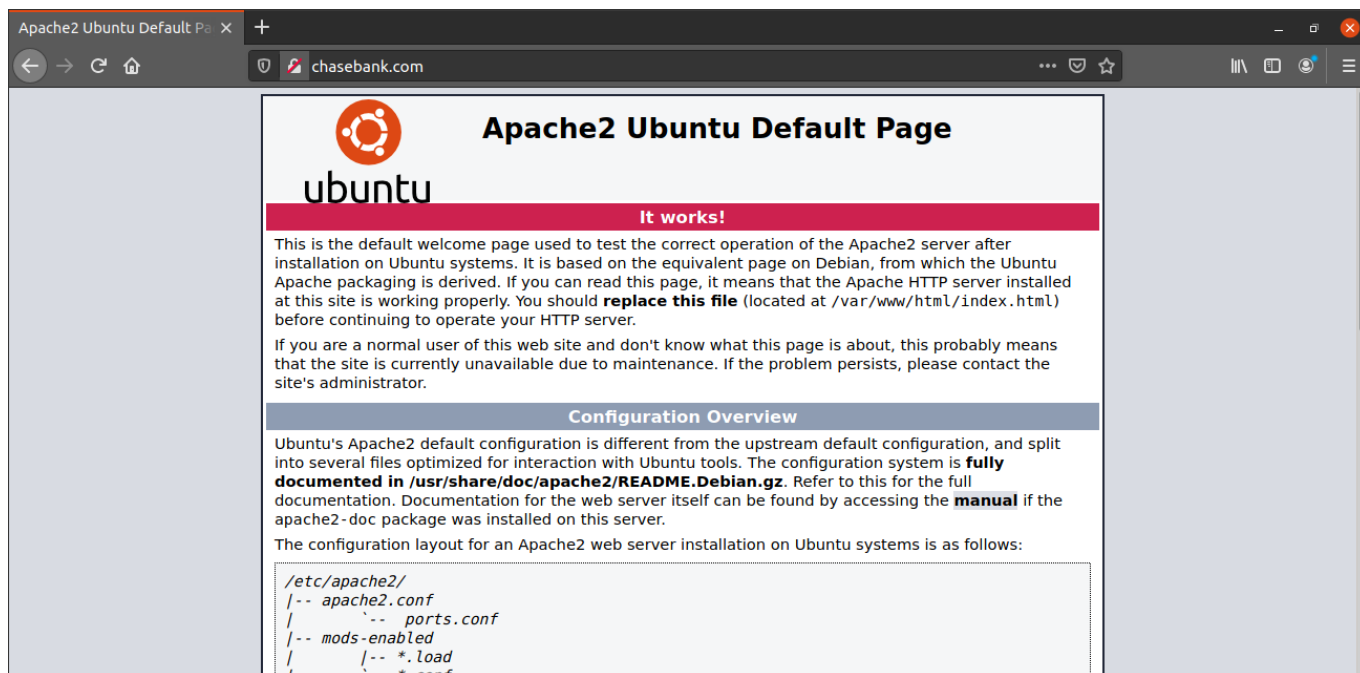
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## Task5 figure 1.3.png



### Decryption:

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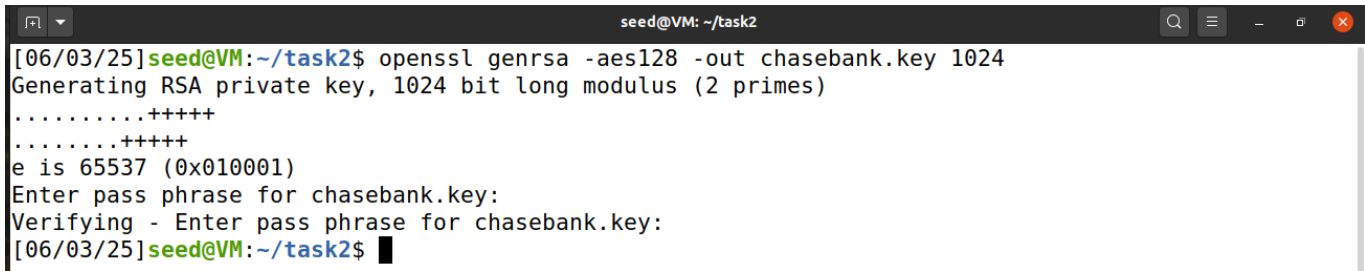
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## Task6 figure 1.1.png



```
seed@VM: ~/task2
[06/03/25]seed@VM:~/task2$ openssl genrsa -aes128 -out chasebank.key 1024
Generating RSA private key, 1024 bit long modulus (2 primes)
.....+++++
.....+++++
e is 65537 (0x010001)
Enter pass phrase for chasebank.key:
Verifying - Enter pass phrase for chasebank.key:
[06/03/25]seed@VM:~/task2$
```

Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task6 figure 1.1.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

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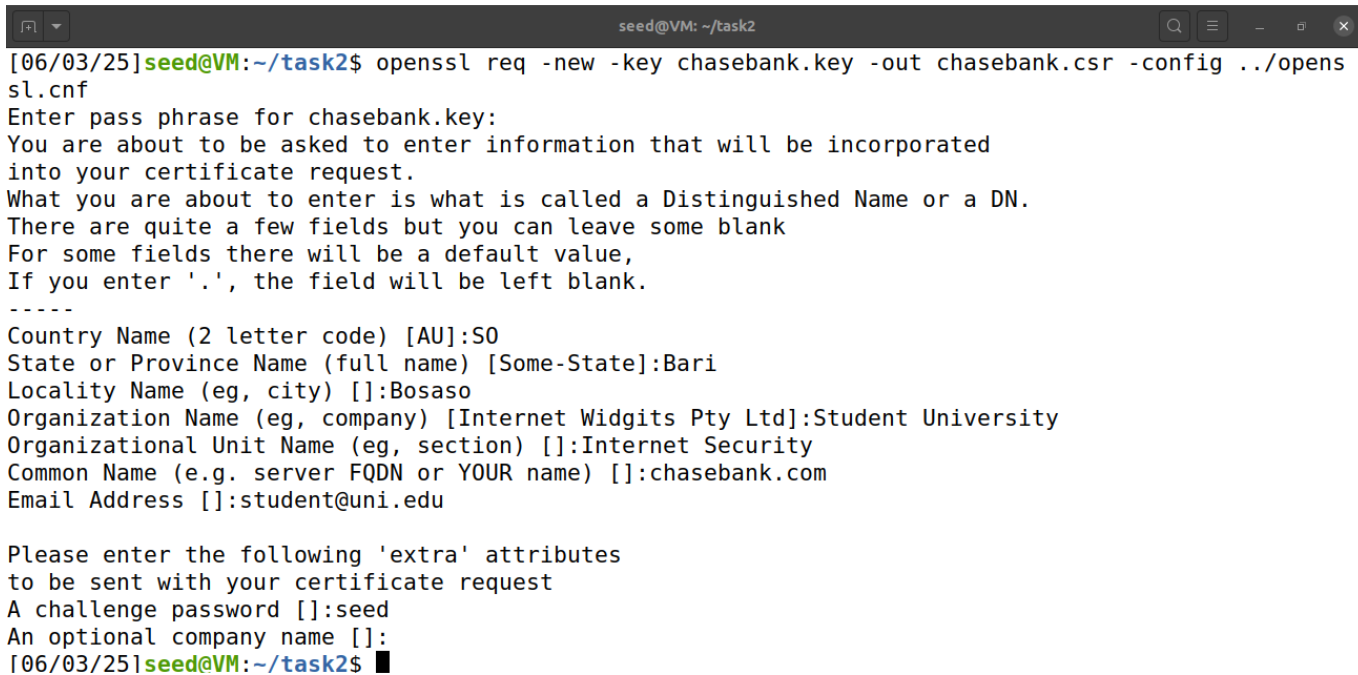
It highlights command execution, result interpretation, and expected outputs.

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## Task6 figure 1.2.png



```
[06/03/25]seed@VM:~/task2$ openssl req -new -key chasebank.key -out chasebank.csr -config ../openssl.cnf
Enter pass phrase for chasebank.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:S0
State or Province Name (full name) [Some-State]:Bari
Locality Name (eg, city) []:Bosaso
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Student University
Organizational Unit Name (eg, section) []:Internet Security
Common Name (e.g. server FQDN or YOUR name) []:chasebank.com
Email Address []:student@uni.edu

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:seed
An optional company name []:
[06/03/25]seed@VM:~/task2$ █
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task6 figure 1.2.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

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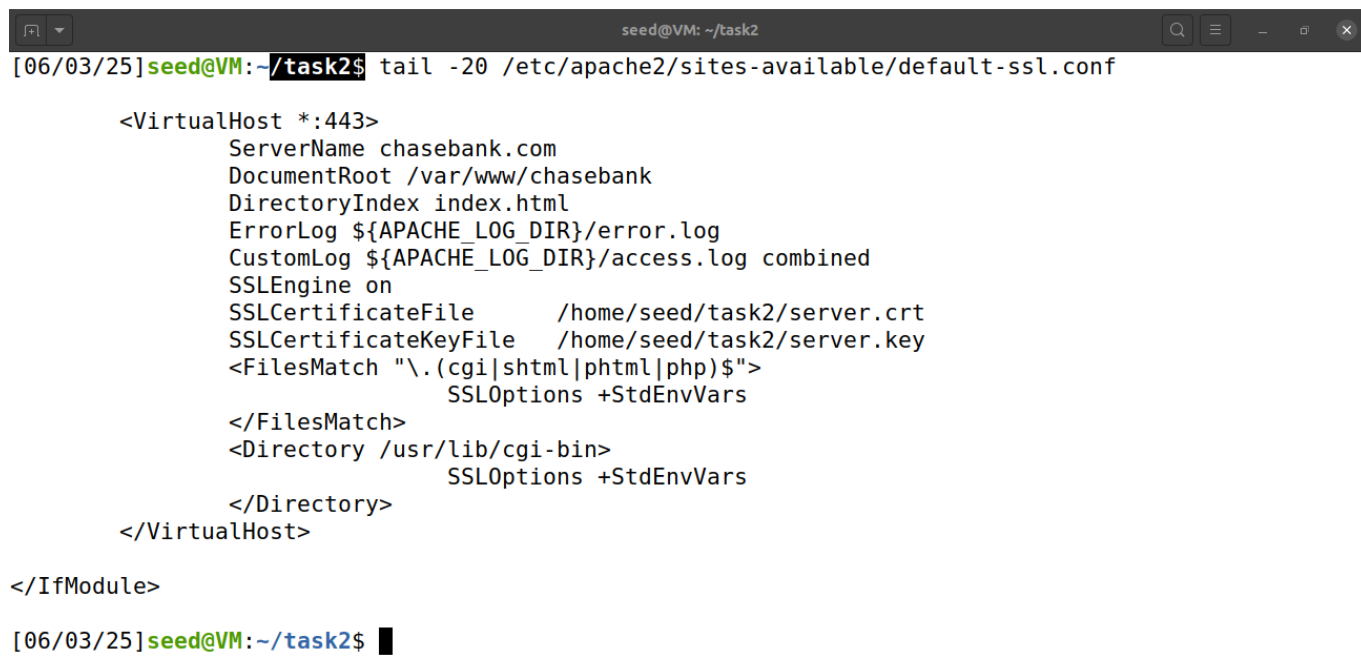
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## Task6 figure 1.3.png



```
seed@VM: ~/task2
[06/03/25] seed@VM:~/task2$ tail -20 /etc/apache2/sites-available/default-ssl.conf
    <VirtualHost *:443>
        ServerName chasebank.com
        DocumentRoot /var/www/chasebank
        DirectoryIndex index.html
        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
        SSLEngine on
        SSLCertificateFile      /home/seed/task2/server.crt
        SSLCertificateKeyFile    /home/seed/task2/server.key
        <FilesMatch "\.(cgi|shtml|phtml|php)$">
            SSLOptions +StdEnvVars
        </FilesMatch>
        <Directory /usr/lib/cgi-bin>
            SSLOptions +StdEnvVars
        </Directory>
    </VirtualHost>
</IfModule>
[06/03/25] seed@VM:~/task2$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'Task6 figure 1.3.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

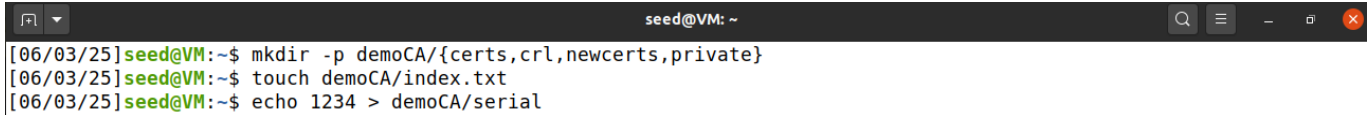
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## first image.png



```
seed@VM: ~  
[06/03/25]seed@VM:~$ mkdir -p demoCA/{certs,crl,newcerts,private}  
[06/03/25]seed@VM:~$ touch demoCA/index.txt  
[06/03/25]seed@VM:~$ echo 1234 > demoCA/serial
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'first image.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

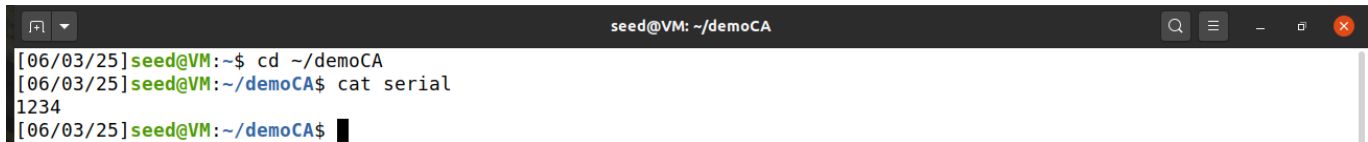
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## second image.png

A screenshot of a Linux terminal window. The title bar at the top reads 'seed@VM: ~/demoCA'. The terminal shows a sequence of commands and their outputs: the first command is 'cd ~/demoCA', followed by 'cat serial', which outputs '1234'. The prompt then returns to the shell.

```
[06/03/25] seed@VM: ~$ cd ~/demoCA
[06/03/25] seed@VM: ~/demoCA$ cat serial
1234
[06/03/25] seed@VM: ~/demoCA$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'second image.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

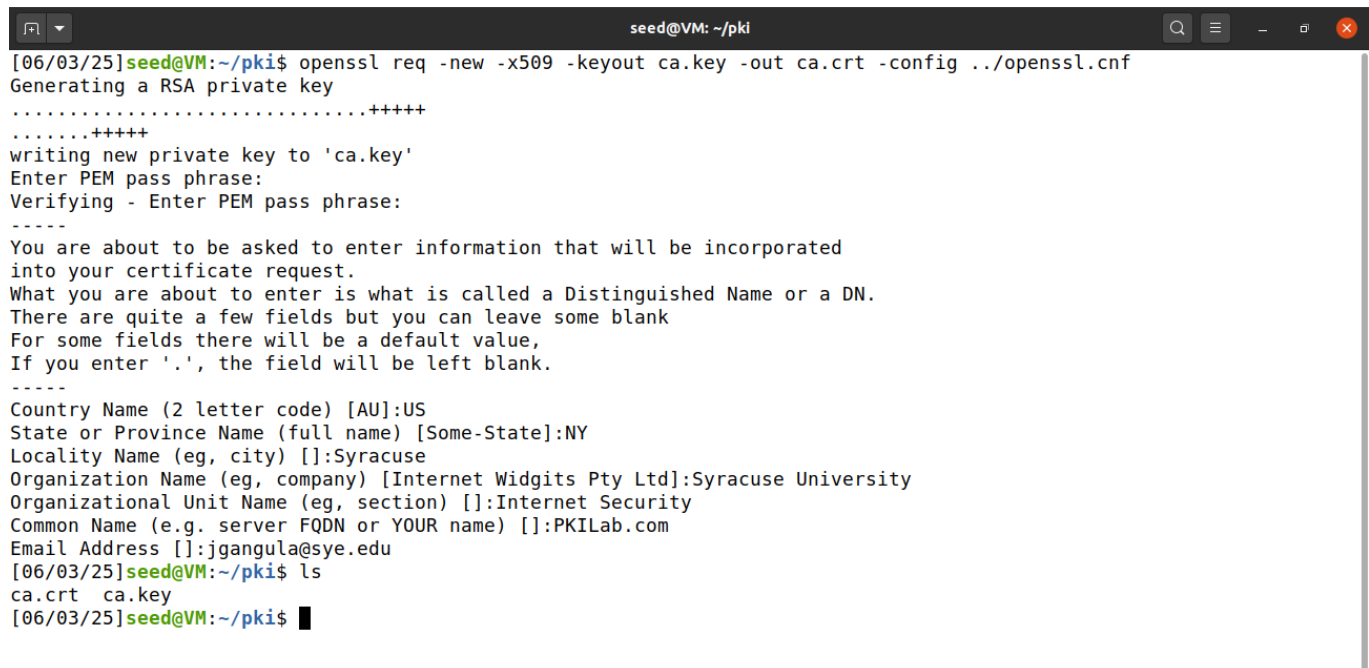
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## task1 p1.1.png



```
seed@VM: ~/pki
[06/03/25]seed@VM:~/pki$ openssl req -new -x509 -keyout ca.key -out ca.crt -config ../openssl.cnf
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'ca.key'
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:NY
Locality Name (eg, city) []:Syracuse
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Syracuse University
Organizational Unit Name (eg, section) []:Internet Security
Common Name (e.g. server FQDN or YOUR name) []:PKILab.com
Email Address []:jgangula@syue.edu
[06/03/25]seed@VM:~/pki$ ls
ca.crt  ca.key
[06/03/25]seed@VM:~/pki$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task1 p1.1.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

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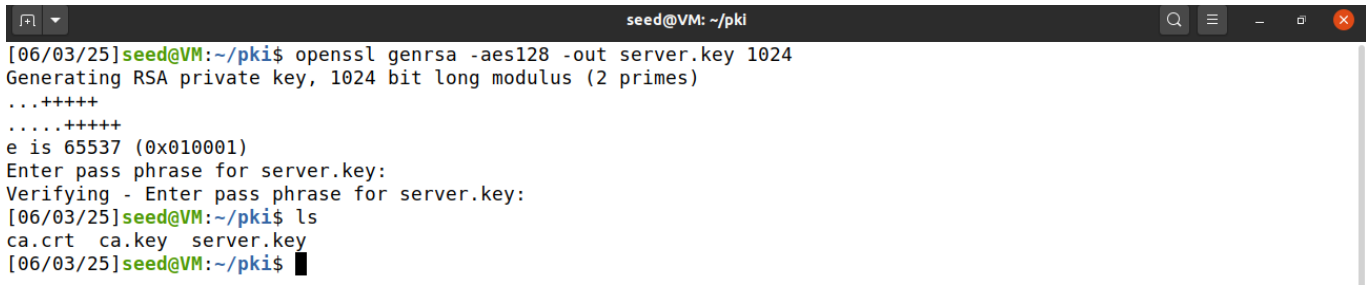
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## task2 p1.1.png



```
seed@VM: ~/pki
[06/03/25]seed@VM:~/pki$ openssl genrsa -aes128 -out server.key 1024
Generating RSA private key, 1024 bit long modulus (2 primes)
...+++++
.....+++++
e is 65537 (0x010001)
Enter pass phrase for server.key:
Verifying - Enter pass phrase for server.key:
[06/03/25]seed@VM:~/pki$ ls
ca.crt  ca.key  server.key
[06/03/25]seed@VM:~/pki$
```

Decryption:

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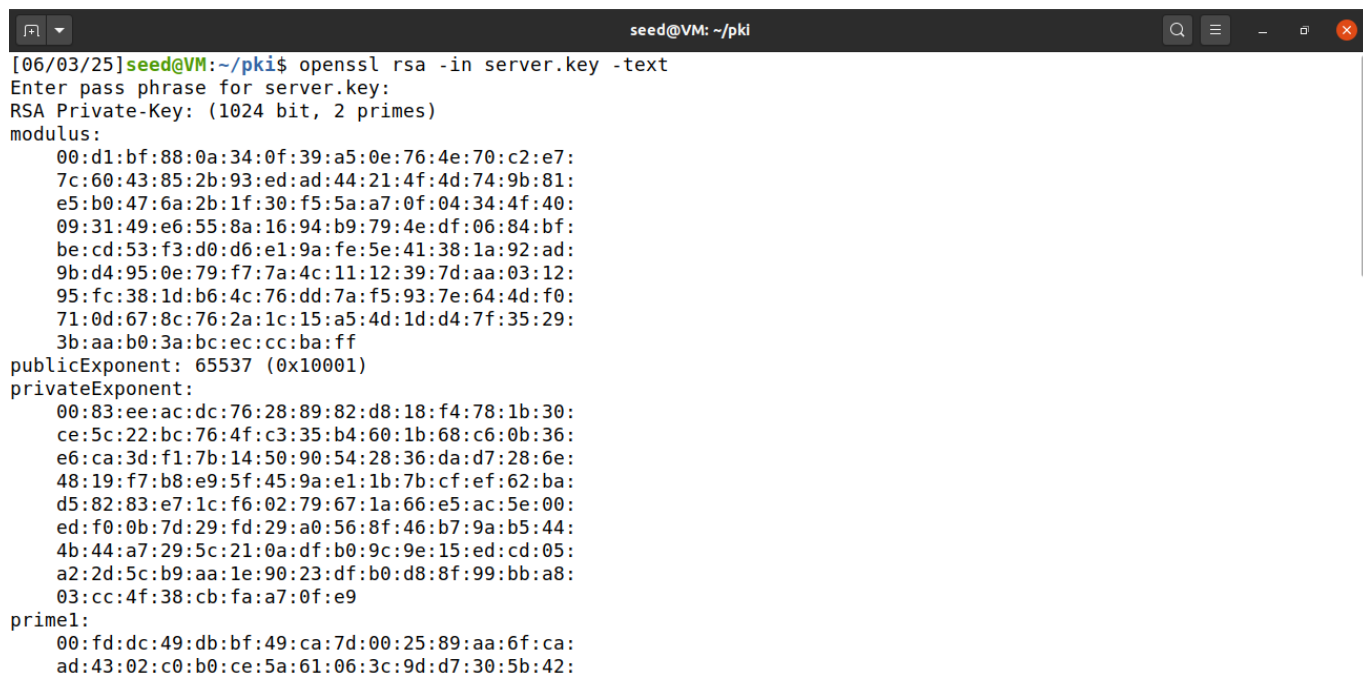
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## task2 p1.2.png



```
[06/03/25]seed@VM:~/pki$ openssl rsa -in server.key -text
Enter pass phrase for server.key:
RSA Private-Key: (1024 bit, 2 primes)
modulus:
 00:d1:bf:88:0a:34:0f:39:a5:0e:76:4e:70:c2:e7:
 7c:60:43:85:2b:93:ed:ad:44:21:4f:4d:74:9b:81:
 e5:b0:47:6a:2b:1f:30:f5:5a:a7:0f:04:34:4f:40:
 09:31:49:e6:55:8a:16:94:b9:79:4e:df:06:84:bf:
 be:cd:53:f3:d0:d6:e1:9a:fe:5e:41:38:1a:92:ad:
 9b:d4:95:0e:79:f7:7a:4c:11:12:39:7d:aa:03:12:
 95:fc:38:1d:b6:4c:76:dd:7a:f5:93:7e:64:4d:f0:
 71:0d:67:8c:76:2a:1c:15:a5:4d:1d:d4:7f:35:29:
 3b:aa:b0:3a:bc:ec:cc:ba:ff
publicExponent: 65537 (0x10001)
privateExponent:
 00:83:ee:ac:dc:76:28:89:82:d8:18:f4:78:1b:30:
 ce:5c:22:bc:76:4f:c3:35:b4:60:1b:68:c6:0b:36:
 e6:ca:3d:f1:7b:14:50:90:54:28:36:da:d7:28:6e:
 48:19:f7:b8:e9:5f:45:9a:e1:1b:7b:cf:ef:62:ba:
 d5:82:83:e7:1c:f6:02:79:67:1a:66:e5:ac:5e:00:
 ed:f0:0b:7d:29:fd:29:a0:56:8f:46:b7:9a:b5:44:
 4b:44:a7:29:5c:21:0a:df:b0:9c:9e:15:ed:cd:05:
 a2:2d:5c:b9:aa:1e:90:23:df:b0:d8:8f:99:bb:a8:
 03:cc:4f:38:cb:fa:a7:0f:e9
prime1:
 00:fd:dc:49:db:bf:49:ca:7d:00:25:89:aa:6f:ca:
 ad:43:02:c0:b0:ce:5a:61:06:3c:9d:d7:30:5b:42:
```

### Decryption:

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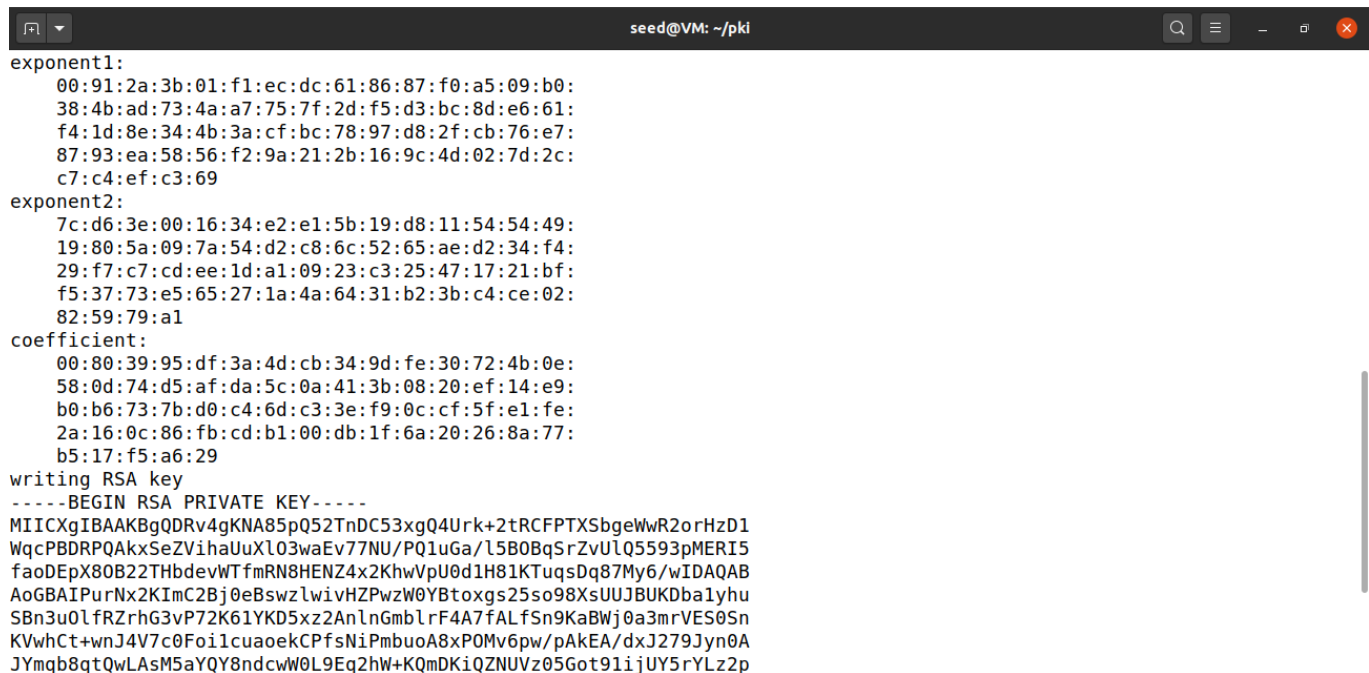
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## task2 p1.3.png

A screenshot of a Linux terminal window with a dark background. The window title is 'seed@VM: ~/pki'. The terminal output shows the generation of RSA keys. It starts with 'exponent1:' followed by a long hexadecimal string. Then 'exponent2:' followed by another long hexadecimal string. Then 'coefficient:' followed by a long hexadecimal string. Finally, it says 'writing RSA key' and '-----BEGIN RSA PRIVATE KEY-----' followed by a very long base64-encoded string.

```
seed@VM: ~/pki
exponent1:
00:91:2a:3b:01:f1:ec:dc:61:86:87:f0:a5:09:b0:
38:4b:ad:73:4a:a7:75:7f:2d:f5:d3:bc:8d:e6:61:
f4:1d:8e:34:4b:3a:cf:bc:78:97:d8:2f:cb:76:e7:
87:93:ea:58:56:f2:9a:21:2b:16:9c:4d:02:7d:2c:
c7:c4:ef:c3:69
exponent2:
7c:d6:3e:00:16:34:e2:e1:5b:19:d8:11:54:54:49:
19:80:5a:09:7a:54:d2:c8:6c:52:65:ae:d2:34:f4:
29:f7:c7:cd:ee:1d:a1:09:23:c3:25:47:17:21:bf:
f5:37:73:e5:65:27:1a:4a:64:31:b2:3b:c4:ce:02:
82:59:79:a1
coefficient:
00:80:39:95:df:3a:4d:cb:34:9d:fe:30:72:4b:0e:
58:0d:74:d5:af:da:5c:0a:41:3b:08:20:ef:14:e9:
b0:b6:73:7b:d0:c4:6d:c3:3e:f9:0c:cf:5f:e1:fe:
2a:16:0c:86:fb:cd:b1:00:db:1f:6a:20:26:8a:77:
b5:17:f5:a6:29
writing RSA key
-----BEGIN RSA PRIVATE KEY-----
MIICXgIBAAKBgQDRv4gKNA85pQ52TnDC53xgQ4Urk+2tRCFPTXSbgeWwR2orHzD1
WqcPBDPRPQAKxSeZVihaUuXl03waEv77NU/PQ1uGa/L5B0BqSrZvU\Q5593pMERI5
faoDEpX80B22THbdevWTfmRN8HENZ4x2KhvVpU0d1H81KTuqsDq87My6/wIDAQAB
AoGBAIPurNx2KImC2Bj0eBswzlwivHZPwzW0YBtoxgs25so98XsUUJBKDbalYhu
SBn3u0lFRZrhG3vP72K61YKD5xz2AnlnGmbLrF4A7fALfSn9KaBWj0a3mrVES0Sn
KVwhCt+wnJ4Y7c0Foi1cuaokCPfsNiPmbuoA8xPOMv6pw/pAkEA/dxJ279Jyn0A
JYmqb8qtQwLAsM5aYQY8ndcw0L9Eq2hW+KQmDKiQZNUVz05Got91ijUY5rYLz2p
```

### Decryption:

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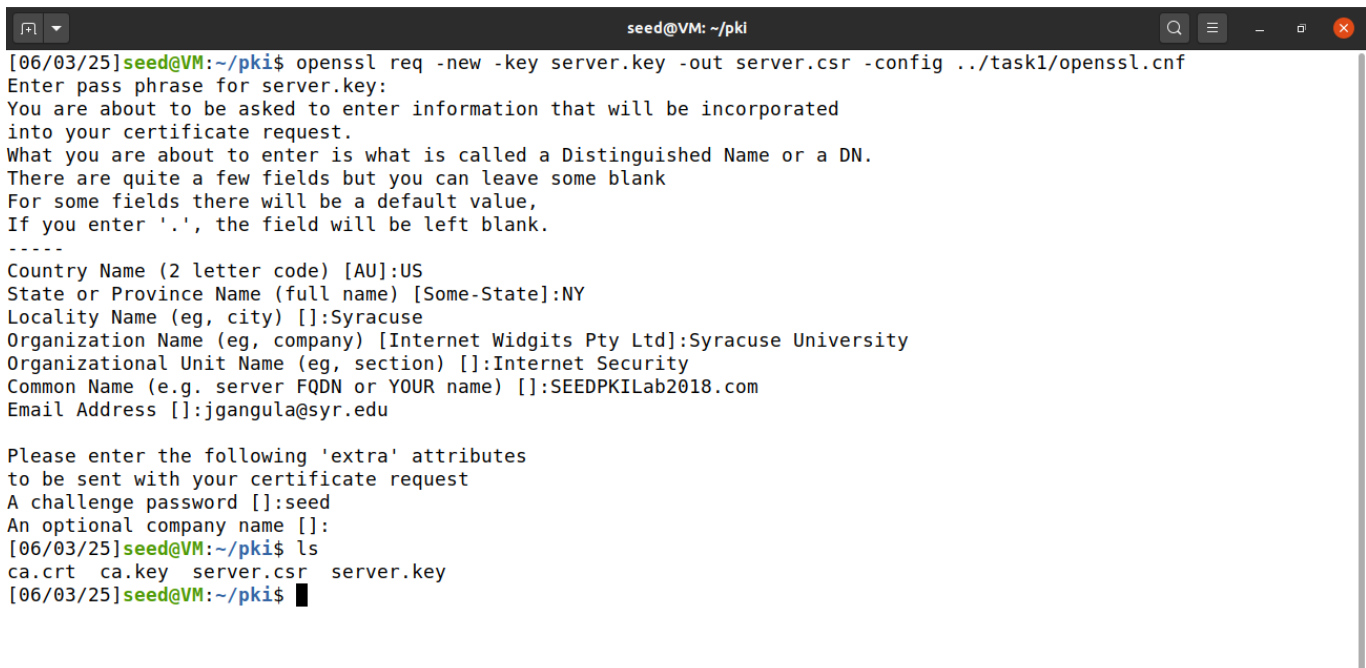
It highlights command execution, result interpretation, and expected outputs.

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## task2 p1.4.png



```
[06/03/25]seed@VM: ~/pki$ openssl req -new -key server.key -out server.csr -config ../task1/openssl.cnf
Enter pass phrase for server.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:NY
Locality Name (eg, city) []:Syracuse
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Syracuse University
Organizational Unit Name (eg, section) []:Internet Security
Common Name (e.g. server FQDN or YOUR name) []:SEEDPKILab2018.com
Email Address []:jgangula@syr.edu

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:seed
An optional company name []:
[06/03/25]seed@VM:~/pki$ ls
ca.crt  ca.key  server.csr  server.key
[06/03/25]seed@VM:~/pki$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task2 p1.4.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

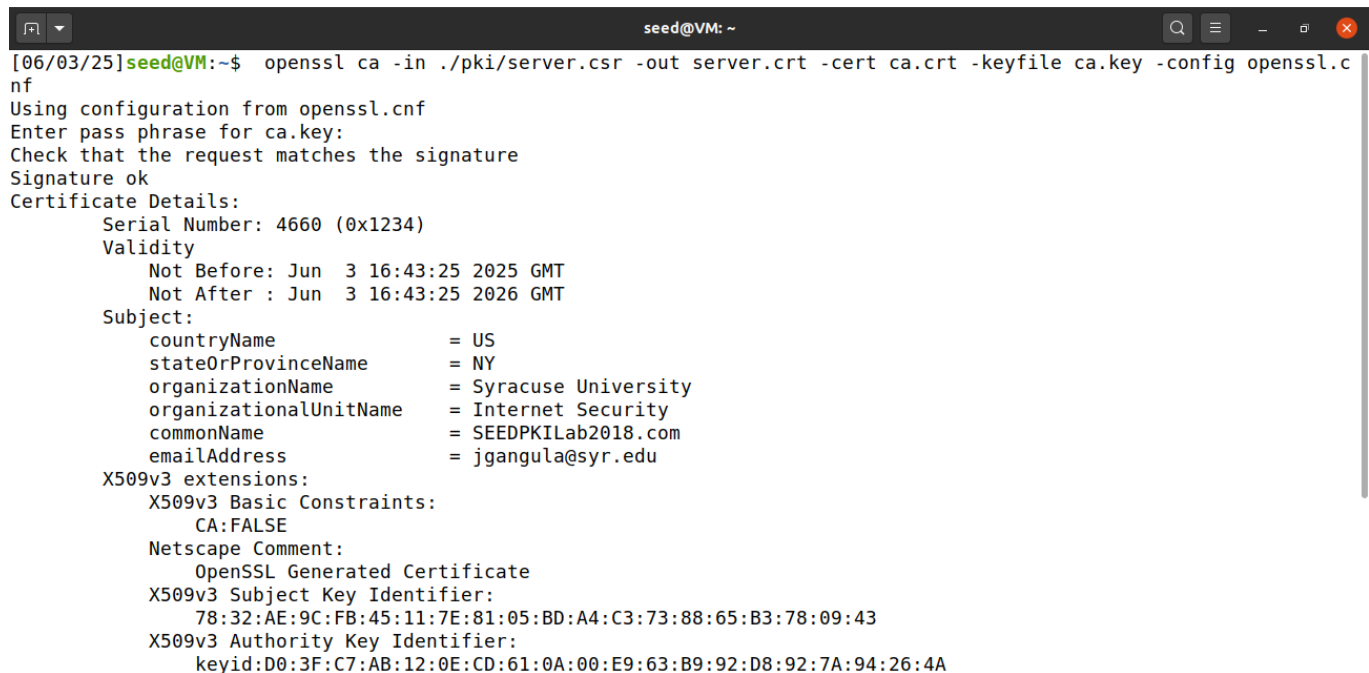
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task3 p1.1.png



```
[06/03/25]seed@VM:~$ openssl ca -in ./pki/server.csr -out server.crt -cert ca.crt -keyfile ca.key -config openssl.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
    Serial Number: 4660 (0x1234)
    Validity
        Not Before: Jun  3 16:43:25 2025 GMT
        Not After : Jun  3 16:43:25 2026 GMT
    Subject:
        countryName           = US
        stateOrProvinceName   = NY
        organizationName      = Syracuse University
        organizationalUnitName = Internet Security
        commonName            = SEEDPKILab2018.com
        emailAddress          = jgangula@syr.edu
    X509v3 extensions:
        X509v3 Basic Constraints:
            CA:FALSE
        Netscape Comment:
            OpenSSL Generated Certificate
        X509v3 Subject Key Identifier:
            78:32:AE:9C:FB:45:11:7E:81:05:BD:A4:C3:73:88:65:B3:78:09:43
        X509v3 Authority Key Identifier:
            keyid:D0:3F:C7:AB:12:0E:CD:61:0A:00:E9:63:B9:92:D8:92:7A:94:26:4A
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.1.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

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This helps reinforce the foundational knowledge required for mastering practical cryptography.

### task3 p1.2.png

```
Certificate is to be certified until Jun  3 16:43:25 2026 GMT (365 days)
Sign the certificate? [y/n]:y

1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Data Base Updated
[06/03/25]seed@VM:~$ █
```

#### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.2.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

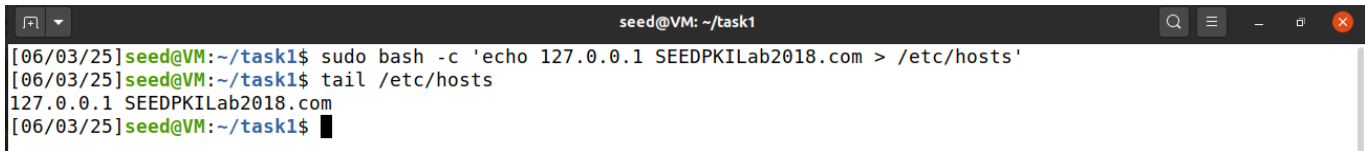
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task3 p1.3.png



```
seed@VM: ~/task1
[06/03/25]seed@VM:~/task1$ sudo bash -c 'echo 127.0.0.1 SEEDPKILab2018.com > /etc/hosts'
[06/03/25]seed@VM:~/task1$ tail /etc/hosts
127.0.0.1 SEEDPKILab2018.com
[06/03/25]seed@VM:~/task1$
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.3.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task3 p1.4.png



```
seed@VM: ~/task3
[06/03/25]seed@VM:~/task3$ cat ../pki/server.key ../pki/server.crt > server.pem
[06/03/25]seed@VM:~/task3$ openssl s_server -cert server.pem -www
Enter pass phrase for server.pem:
Using default temp DH parameters
ACCEPT
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.4.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

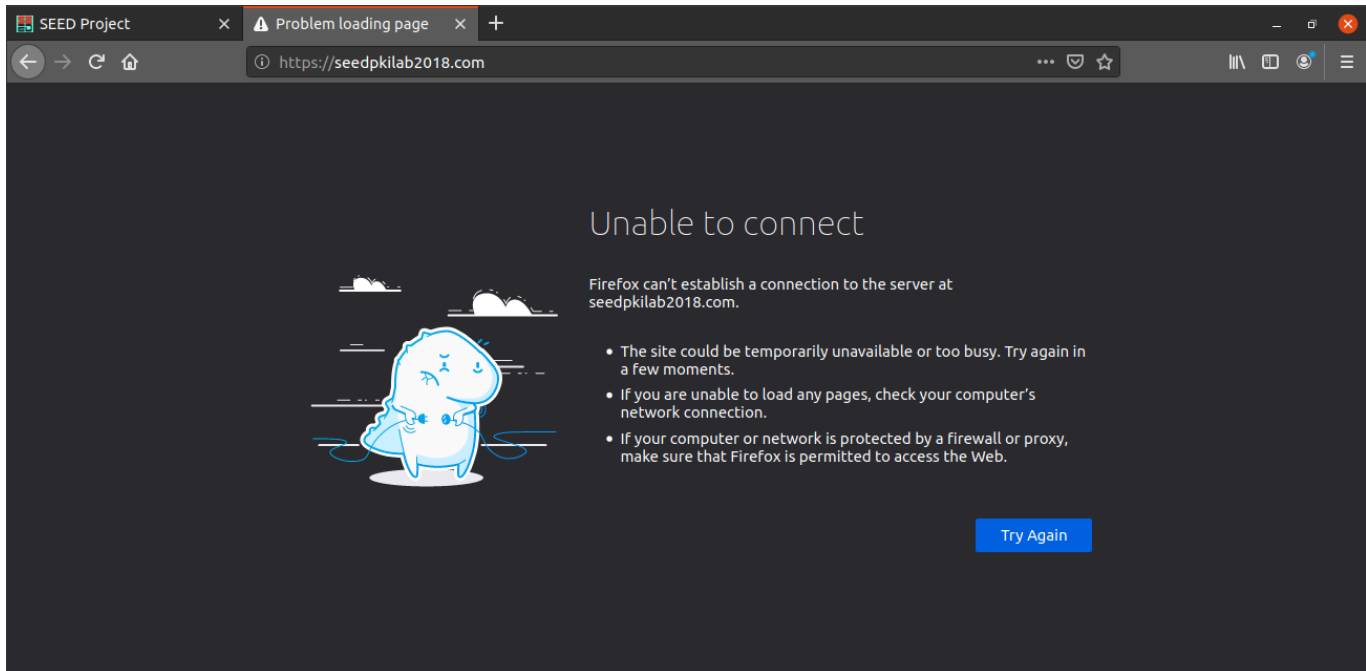
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task3 p1.5.png



### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.5.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

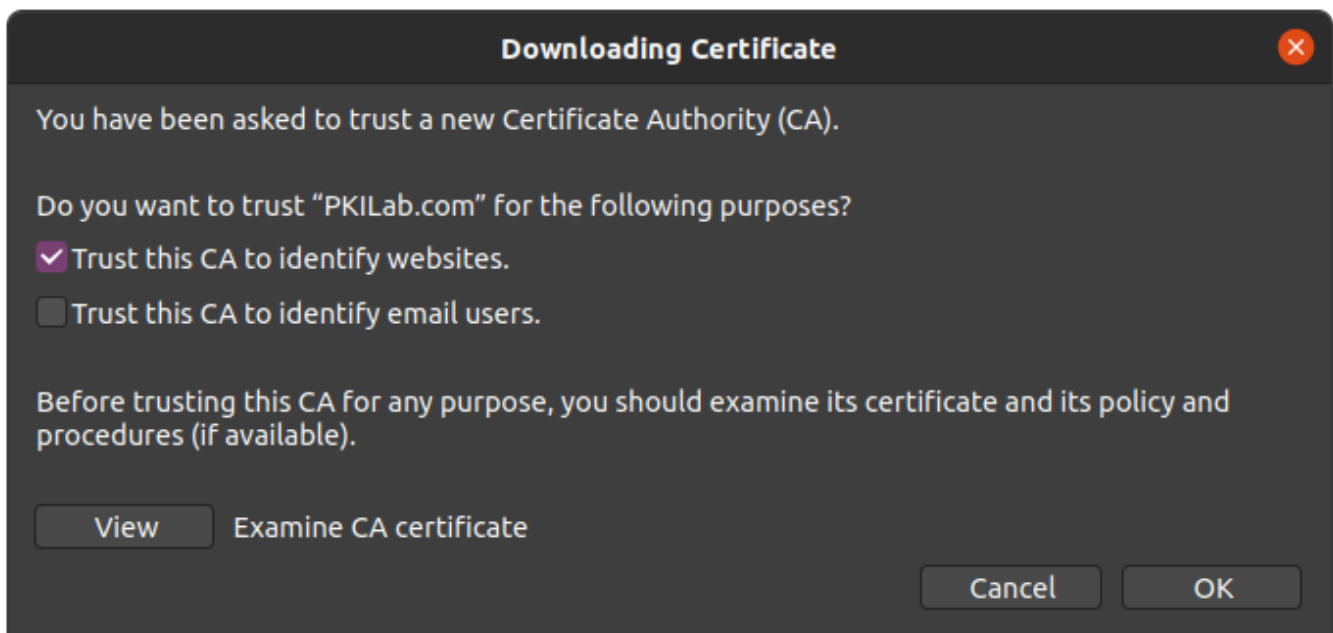
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

### task3 p1.6.png



Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.6.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

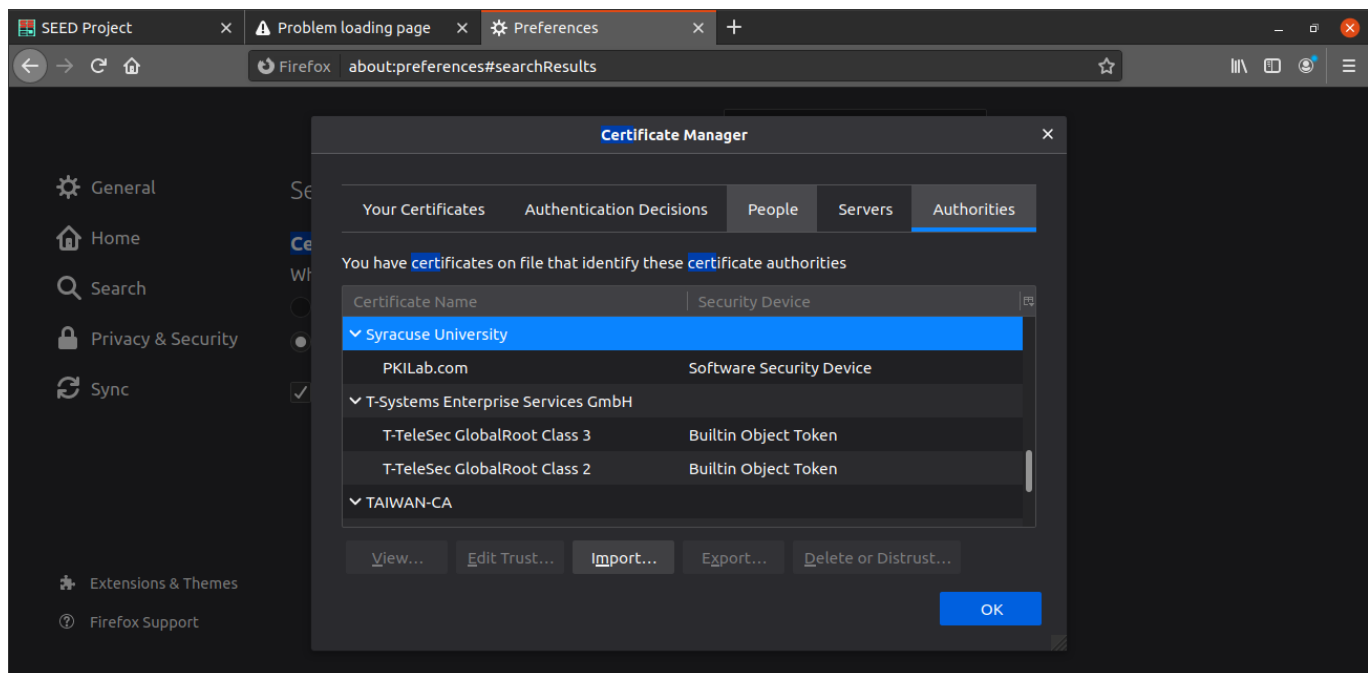
Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task3 p1.7.png



### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.7.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

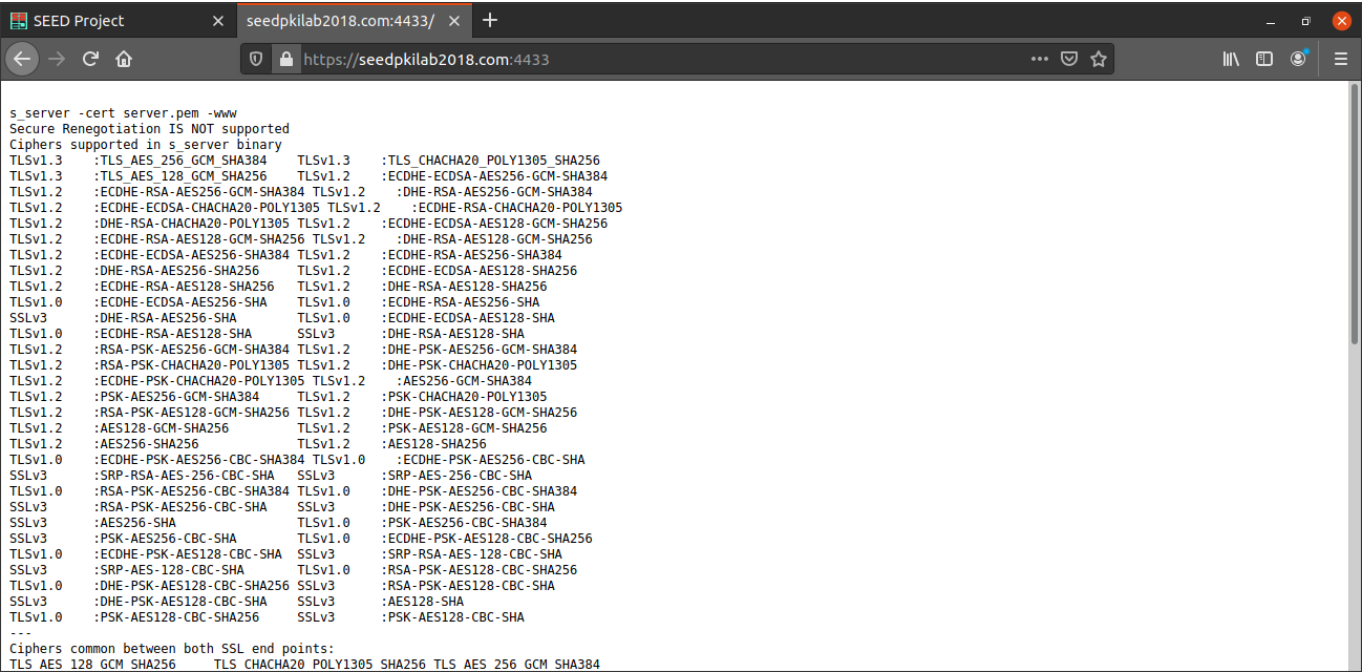
It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.



task3 p1.8.png



Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task3 p1.8.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

Each screenshot captures a moment critical to understanding the underlying mechanism of operating system functionality in cryptographic tasks.

It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.

## task4 p1.1.png

```
[06/03/25]seed@VM:~$ tail -20 /etc/apache2/sites-available/default-ssl.conf

<VirtualHost *:443>
    ServerName chasebank.com
    DocumentRoot /var/www/chasebank
    DirectoryIndex index.html
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
    SSLEngine on
    SSLCertificateFile      /home/seed/task2/server.crt
    SSLCertificateKeyFile   /home/seed/task2/server.key
    <FilesMatch "\.(cgi|shtml|phtml|php)$">
        SSLOptions +StdEnvVars
    </FilesMatch>
    <Directory /usr/lib/cgi-bin>
        SSLOptions +StdEnvVars
    </Directory>
</VirtualHost>

</IfModule>
```

### Decryption:

This figure is part of Lab 4 under the Cryptography course. It illustrates a key step in demonstrating operating system-level commands, configurations, or output verifications.

The image titled 'task4 p1.1.png' shows specific actions performed by the student using Linux terminal or system GUI tools.

These tasks often reflect file system interaction, process handling, permissions management, or encryption behavior.

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It highlights command execution, result interpretation, and expected outputs.

By analyzing this step, we understand how system-level features contribute to security and control in a computing environment.

This helps reinforce the foundational knowledge required for mastering practical cryptography.